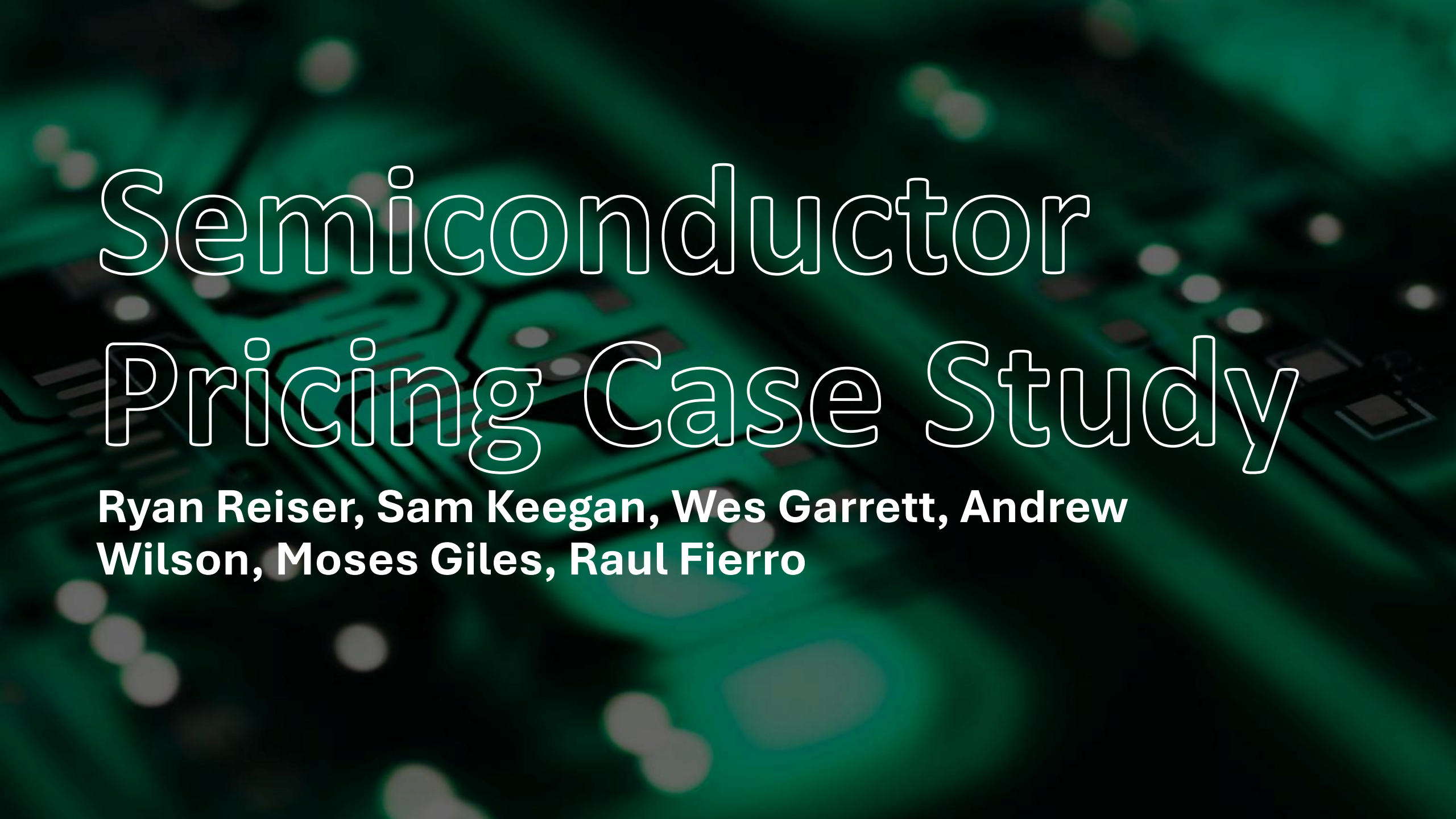


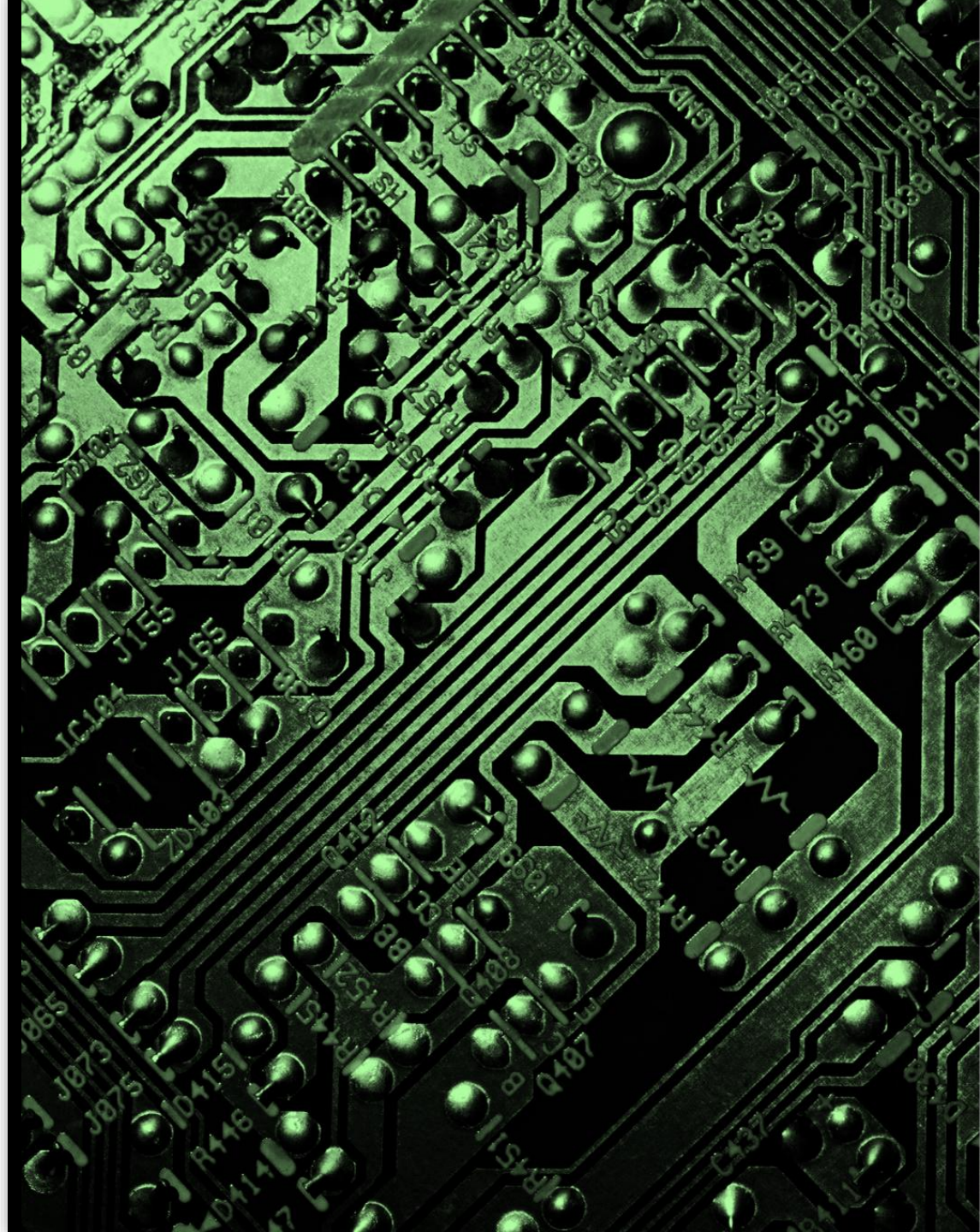


Semiconductor Pricing Case Study

**Ryan Reiser, Sam Keegan, Wes Garrett, Andrew
Wilson, Moses Giles, Raul Fierro**

Background

- Semiconductors are the foundation of microchips, and are used in just about every modern electronic device
- Majority of semiconductors are manufactured internationally
- Dynamics of the semiconductor market will help build better understanding of tech markets
- Economic instability and policy change is impacting production and costs



Modeling Approach

- Run basic regressions to test initial relationships between variables
- Include lagged semiconductor trade stats to predict:
 - Semiconductor Prices
- Apply log transformations for linearization
- Use differences of logs to ensure stationarity before ARIMA
- Evaluate models with:
 - Adjusted R^2 , AIC → overall performance
 - Z-stats, signs of coefficients → variable significance
- Generate forecasts for employment & unemployment

Question?

How can historical revenue of the global semiconductor industry be used to predict future revenues?

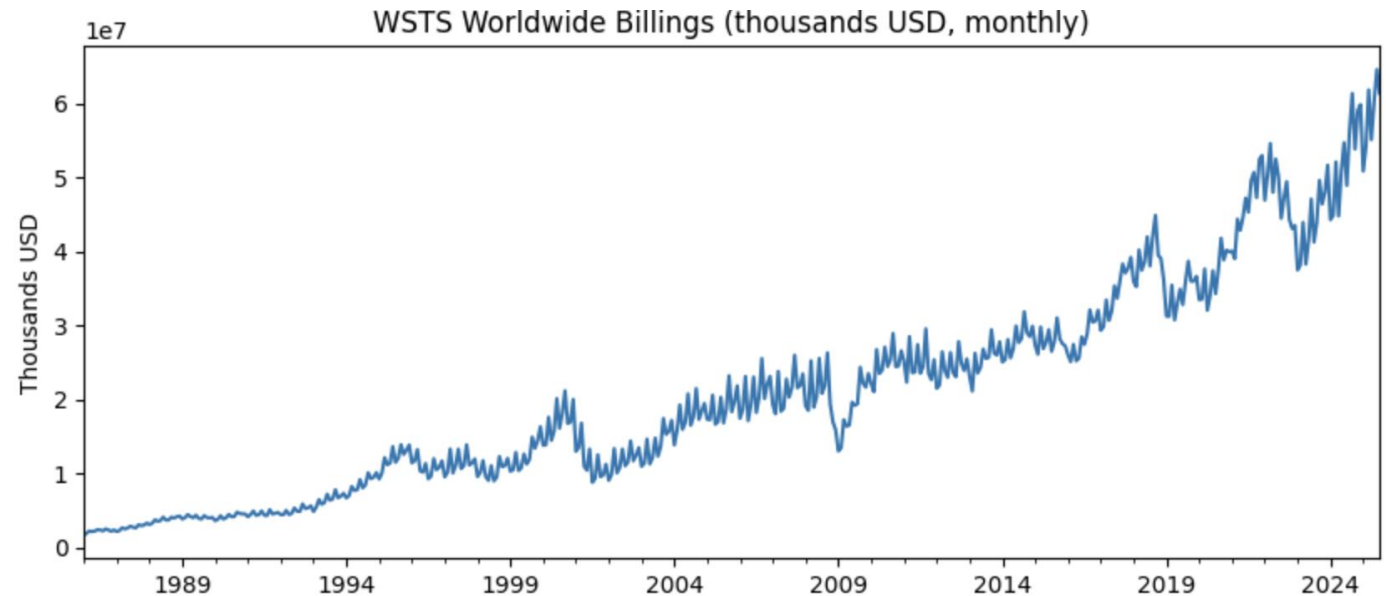
Data:

- World Semiconductor Trade Statistics
- Time Frame: 1980–2024 quarterly
- Transformations:
 - Log transformation
 - Differencing for stationarity
- Link semiconductor revenue changes to employment shifts in the tech sector.
- Build reliable forecasts of semiconductor growth over time.

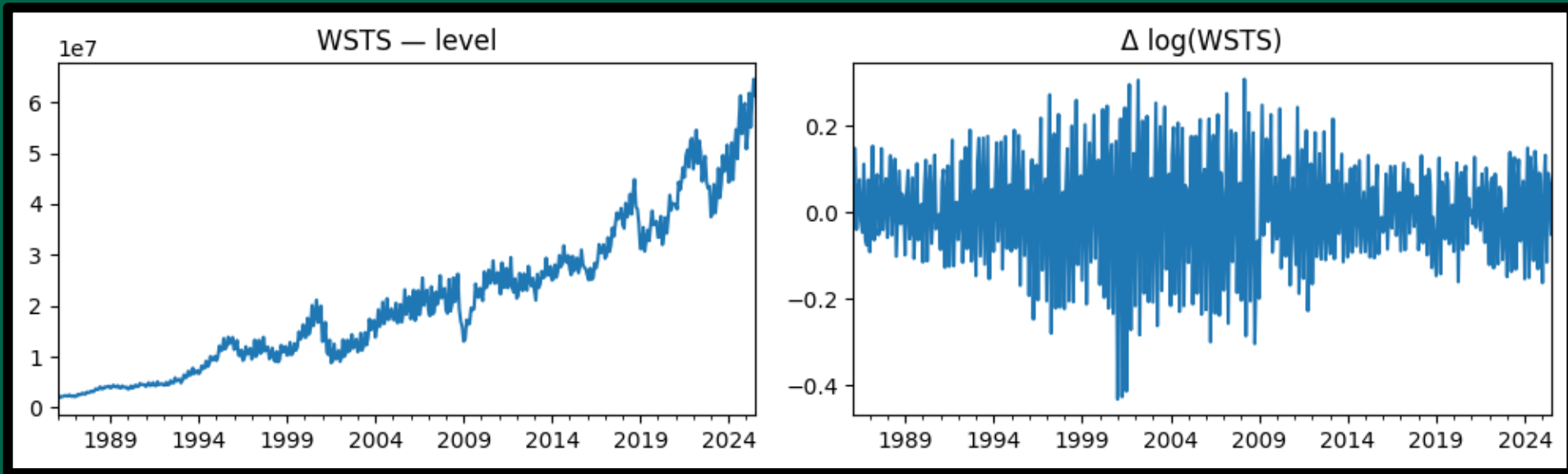


Plotting the Time Series Data

- WSTS data shows monthly global semiconductor billings.
- Clear long term upward trend since the late 1980s, with cyclical dips.
- Sharp growth in recent years driven by AI, consumer electronics, and automotive demand.
- Illustrates both the expansion and volatility of the semiconductor industry.



Preparing the Time Series



Monthly WSTS billings (1988–2024) converted to **month-end frequency**

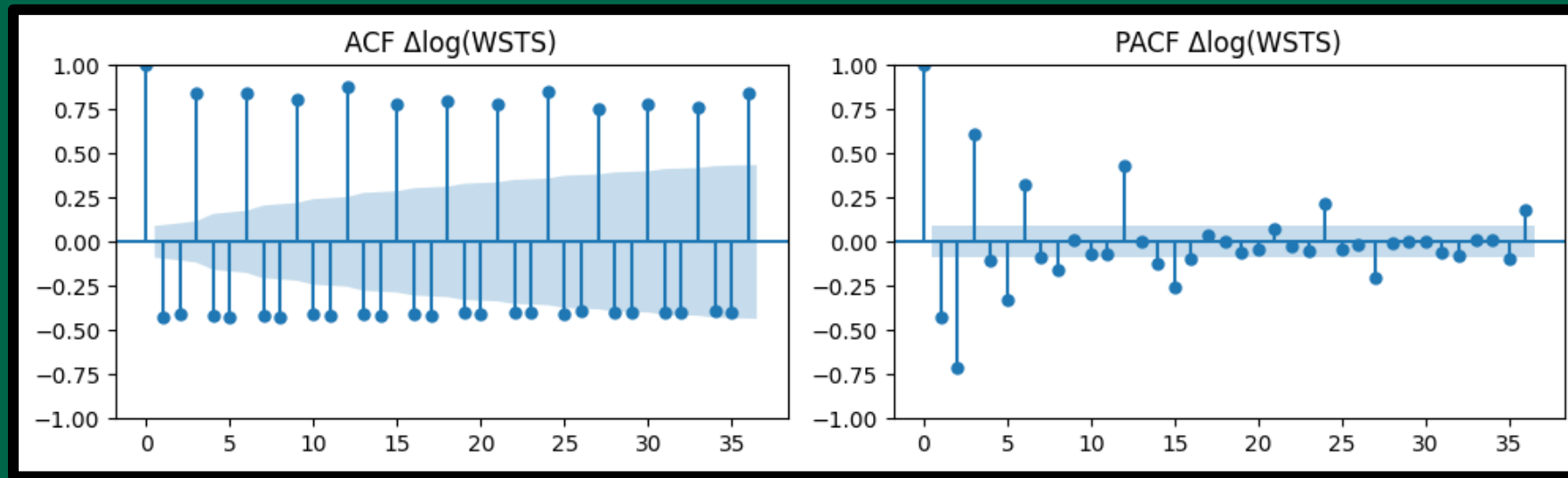
Log transform applied, stabilizes variance

First difference of log values, monthly % change

Plots:

- Left: long-term growth trend
- Right: short-term volatility

WSTS Trends



Monthly % changes ($\Delta\log$) made the data stable over time

ACF:

- Repeating ups and downs
- Signals short-term cycles in semiconductor sales

PACF:

- Few strong early spikes
- Recent months drive most of the predictability

Semiconductor sales move in short, repeating cycles

OLS Quadratic Trend

- Quadratic model is statistically significant which is reflected in R^2
- This model tells us that semiconductor billings have been increasing on average by 1.02% every quarter
- The growth rate has been declining

OLS Regression Results

Dep. Variable:

y

R-squared:

0.952

Model:

OLS

Adj. R-squared:

0.952

Method:

Least Squares

F-statistic:

4655.

Date:

Sat, 11 Oct 2025

Prob (F-statistic):

2.09e-311

Time:

20:31:17

Log-Likelihood:

122.26

No. Observations:

475

AIC:

-238.5

Df Residuals:

472

BIC:

-226.0

Df Model:

2

Covariance Type:

nonrobust

coef

std err

t

P>|t|

[0.025

0.975]

const

14.8074

0.026

575.667

0.000

14.757

14.858

x1

0.0102

0.000

40.542

0.000

0.010

0.011

x2

-8.867e-06

5.12e-07

-17.319

0.000

-9.87e-06

-7.86e-06

Omnibus:

16.055

Durbin-Watson:

0.477

Prob(Omnibus):

0.000

Jarque-Bera (JB):

16.942

Skew:

0.419

Prob(JB):

0.000209

Kurtosis:

3.390

Cond. No.

3.01e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly speci

[2] The condition number is large, 3.01e+05. This might indicate that there are strong multicollinearity or other numerical problems.

In-sample accuracy (levels):

Linear trend

MAE=3,065,615

RMSE=4,164,079

MAPE=19.06%

(thousands USD)

Quadratic trend

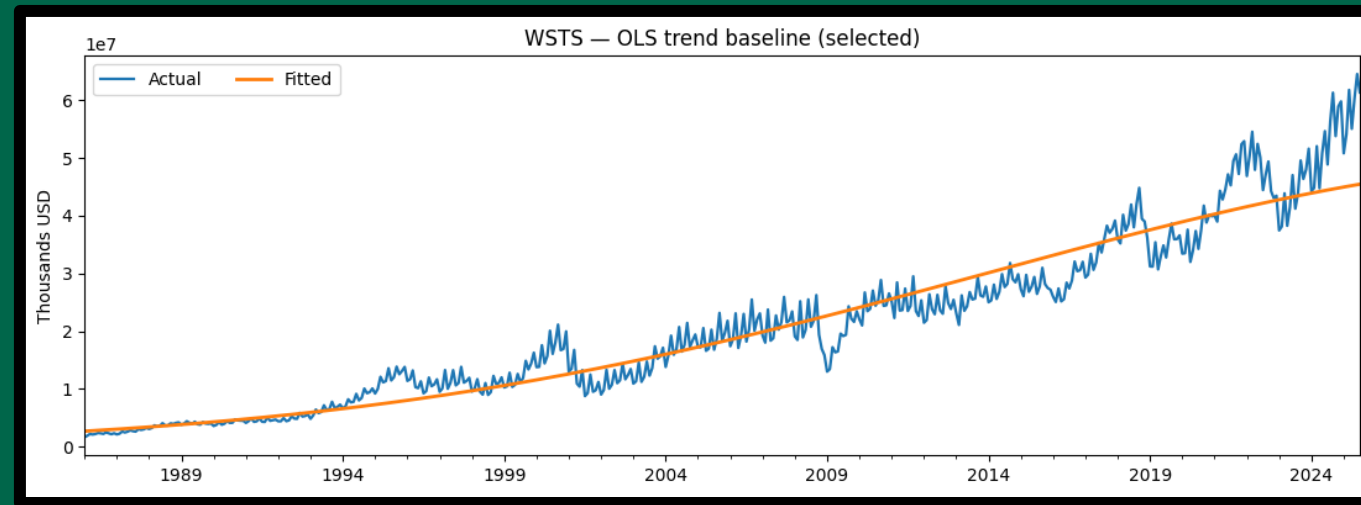
MAE=2,812,896

RMSE=4,052,863

MAPE=14.58%

(thousands USD)

Actual vs Fitted



- The quadratic model explains the overall trend decently well, but misses out on short term dips such as 2008
- The model underestimates the recent growth in the model

Baseline ARIMA Model - Forecasting Semiconductor Revenue (WSTS)

We tested a range of AR and MA terms to find the best-fitting model.

Selected Model: ARIMA(2,1,4) — best overall performance.

AIC: -1384.55 BIC: -1355.42

All coefficients were statistically significant ($|z| > 2$, $p < 0.05$).

We can reject the null hypothesis, confirming the model terms are meaningful.

Residuals show no autocorrelation (Ljung–Box $p > 0.05$) $\rightarrow$ model fits the data well.

Captures both long-term trends and short-term cycles in semiconductor revenue.

```

=====
SARIMAX Results
=====
Dep. Variable:      wsts_world_billings_thous      No. Observations:      475
Model:              ARIMA(2, 1, 4)                Log Likelihood         699.275
Date:              Sat, 11 Oct 2025                AIC                   -1384.550
Time:              20:01:28                        BIC                   -1355.421
Sample:            01-31-1986                      HQIC                  -1373.094
                  - 07-31-2025
Covariance Type:   opg
=====

```

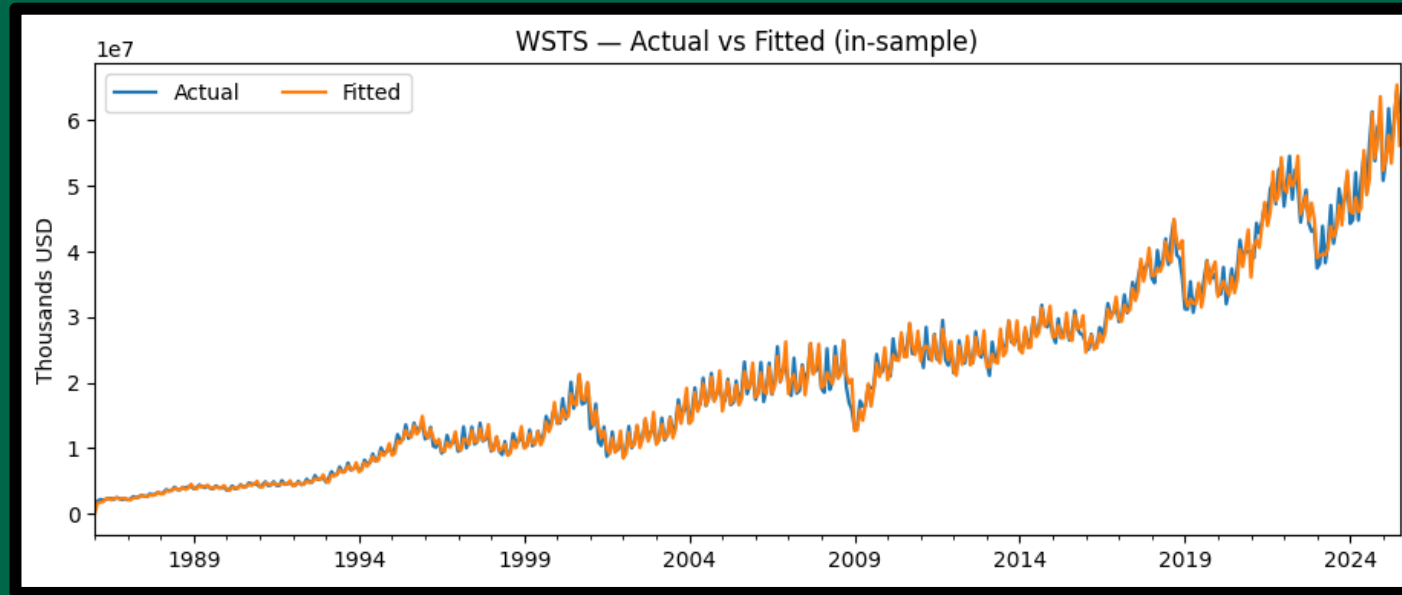
	coef	std err	z	P> z	[0.025	0.975]
ar.L1	-0.9984	0.004	-274.748	0.000	-1.006	-0.991
ar.L2	-0.9963	0.002	-439.184	0.000	-1.001	-0.992
ma.L1	0.9403	0.043	22.033	0.000	0.857	1.024
ma.L2	1.0060	0.059	17.118	0.000	0.891	1.121
ma.L3	0.1696	0.060	2.847	0.004	0.053	0.286
ma.L4	0.1578	0.047	3.346	0.001	0.065	0.250
sigma2	0.0030	0.000	18.864	0.000	0.003	0.003

```

=====
Ljung-Box (L1) (Q):      0.19      Jarque-Bera (JB):      104.30
Prob(Q):                 0.66      Prob(JB):              0.00
Heteroskedasticity (H):  0.75      Skew:                  -0.68
Prob(H) (two-sided):     0.07      Kurtosis:              4.85
=====
Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

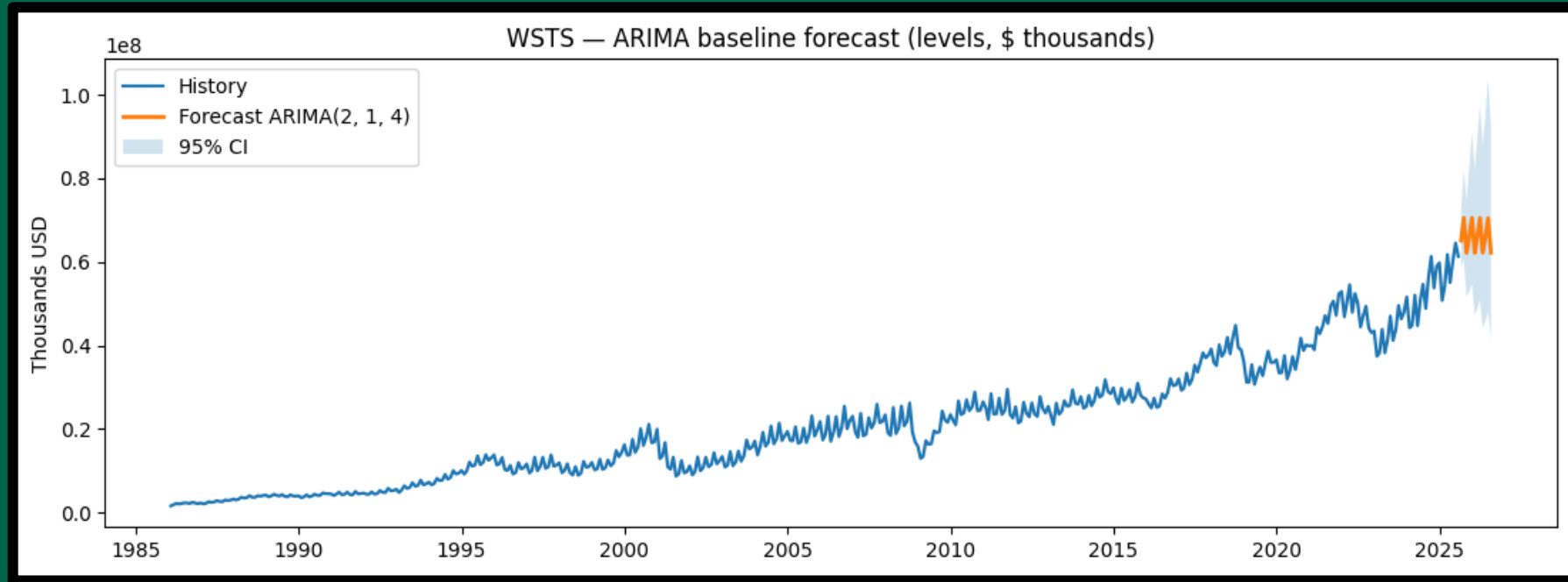
```

ARIMA Baseline Model



- The ARIMA(2, 1, 4) model closely tracks observed WSTS semiconductor billings.
- Captures both long-term growth and short-term cyclical movements.
- Small deviations occur during rapid demand spikes or global slowdowns.
- Produces strong in-sample accuracy with a MAPE around 4.5%.
- Provides a solid foundation for reliable future forecasts.

ARIMA Forecast



- Forecast extends the WSTS semiconductor billings into the next 12 months.
- ARIMA(2,1,4) model predicts continued upward growth with normal short-term fluctuations.
- Confidence interval widens slightly, showing moderate forecast uncertainty.
- Indicates sustained demand driven by AI, consumer electronics, and automotive sectors.
- Confirms the semiconductor industry's resilient long term expansion trend.

VAR Model – Testing Relationship Between Semiconductor Revenue and Prices

Goal: Test if semiconductor prices (IPPI) help predict semiconductor revenue (WSTS).

Model: VAR(12) - chosen using AIC = -16.41 (best fit).

Key Stats:

- BIC = -15.38

Findings:

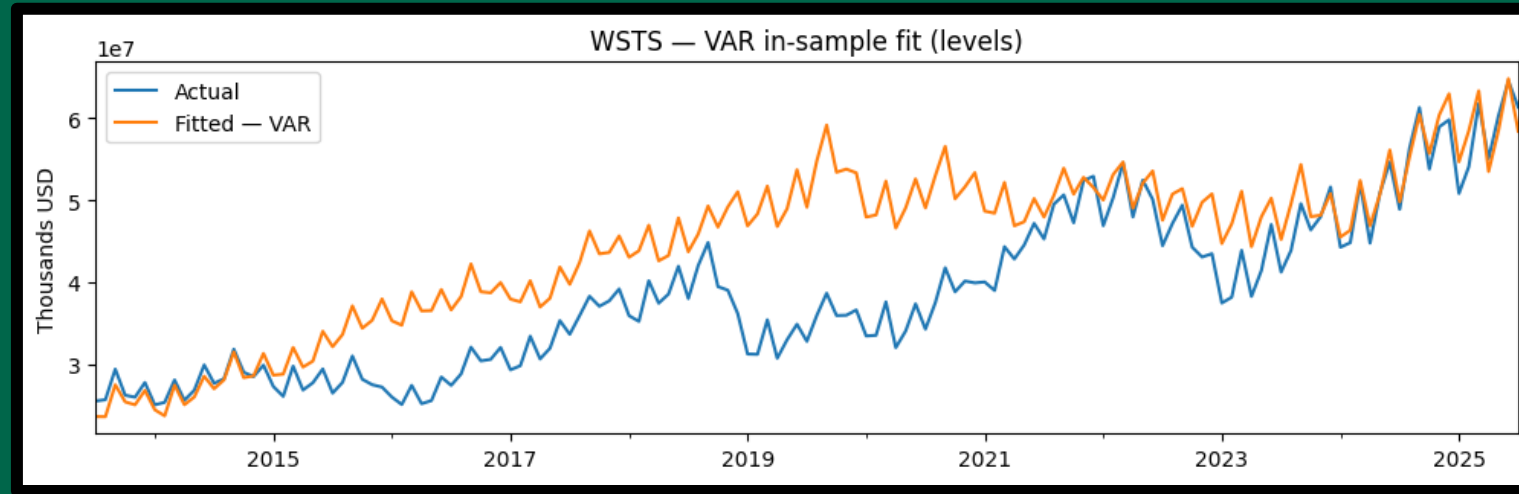
- Most p-values > 0.05 , We fail to reject the null hypothesis.
- IPPI is not statistically significant in predicting WSTS.

Interpretation:

WSTS revenue is mostly driven by its own history, not short-term price changes.

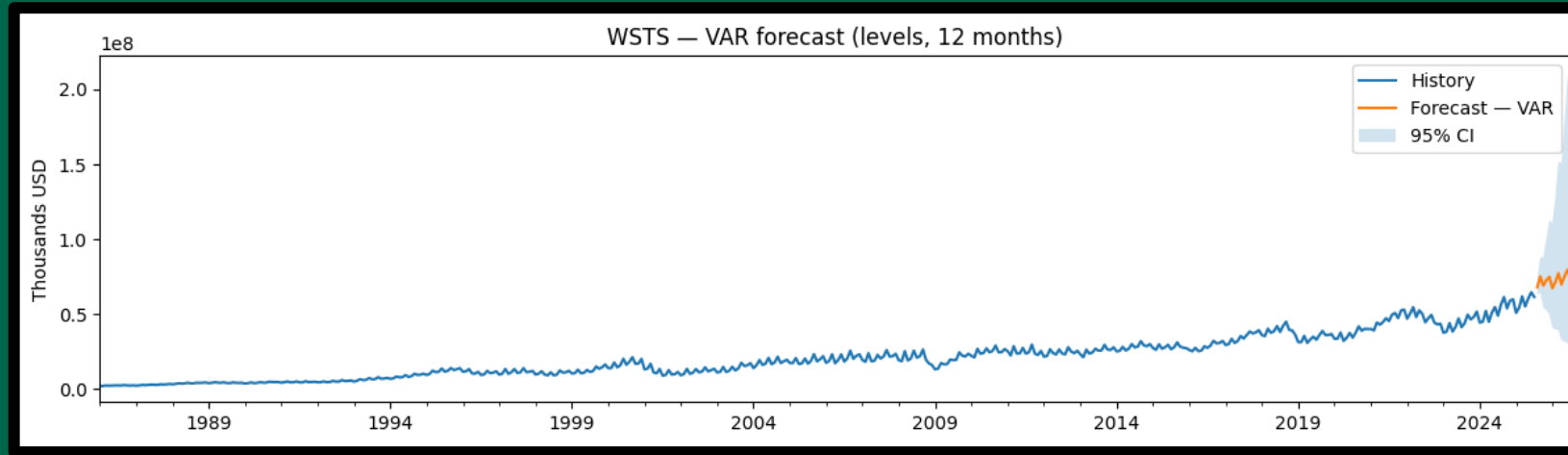
```
=== VAR Model Summary ===
Summary of Regression Results
=====
Model:                      VAR
Method:                     OLS
Date:                      Sun, 12, Oct, 2025
Time:                      18:49:45
-----
No. of Equations:          2.00000    BIC:                      -15.3824
Nobs:                     145.000    HQIC:                     -15.9918
Log likelihood:           828.151    FPE:                      7.52906e-08
AIC:                      -16.4089    Det(Omega_mle):          5.47745e-08
=====
```

VAR Model (In-Sample Fit)



- Compared actual vs fitted semiconductor revenue (WSTS).
- The model tracks general trends but overestimates peaks.
- Suggests the VAR can explain broad movements but misses finer cycles.
- Confirms that most variation in revenue comes from WSTS's own past.
- External price index (IPPI) adds little predictive improvement.

VAR Forecast (12 Months Ahead)



- Forecast extends 12 months beyond the data.
- Trend shows moderate continued growth in global semiconductor revenue.
- Confidence interval widens, then increased uncertainty further out.
- VAR results confirm: revenue depends mainly on its own momentum.
- IPPI impact remains weak or statistically insignificant.

Bonus: Predicting Taiwanese GDP

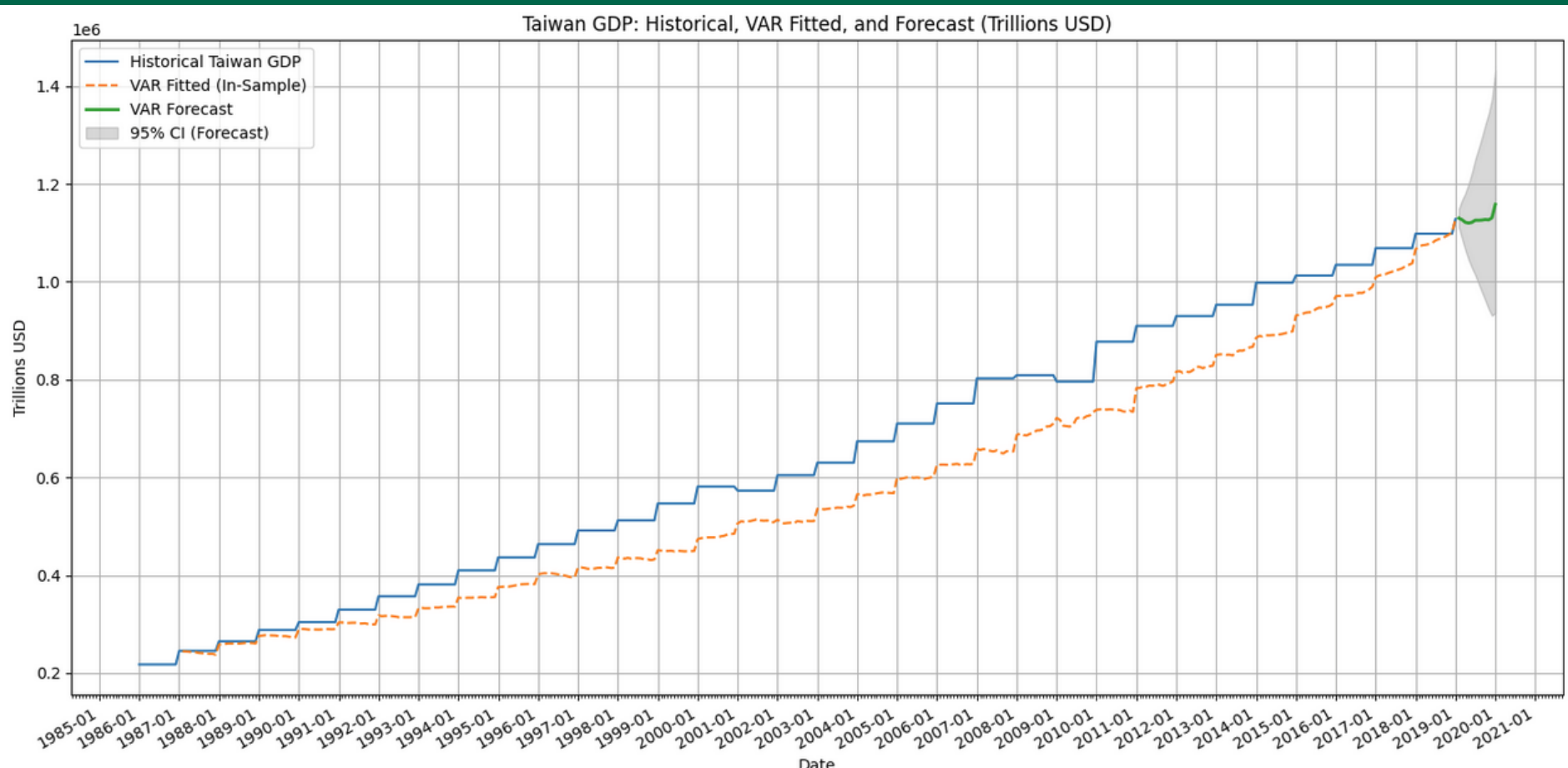
- Taiwan manufactures ~50% of global semiconductors
- Historical Taiwanese GDP from FRED

```
=== Fitted VAR Model Summary ===
Summary of Regression Results
=====
Model:                VAR
Method:               OLS
Date:                Tue, 14, Oct, 2025
Time:                04:20:35
-----
No. of Equations:      2.00000    BIC:                -14.9190
Nobs:                 384.000    HQIC:               -15.2294
Log likelihood:       1923.47    FPE:                1.98391e-07
AIC:                  -15.4334    Det(Omega_mle):     1.74879e-07
-----
Results for equation gdp_tw_log_diff
=====
```

	coefficient	std. error	t-stat	prob
const	0.003402	0.000913	3.727	0.000
L1.gdp_tw_log_diff	-0.045527	0.036588	-1.244	0.213
L1.wsts_world_log_diff	-0.030293	0.009080	-3.336	0.001
L2.gdp_tw_log_diff	0.011130	0.036338	0.306	0.759
L2.wsts_world_log_diff	0.007965	0.009190	0.867	0.386
L3.gdp_tw_log_diff	-0.049589	0.037094	-1.337	0.181

Results

- Data only goes to 2019, and only has 1 data point per year
- Generally correct prediction for upward growth



Predicting Semiconductor Sales

- Very accurate to reality

